Practical Screening of Green Materials Synthesized via Knoevenagel Condensation for Application in Organic Solar Cells

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1 Overview

Organic Solar Cell (OSC) research has experienced incredible advancements over the previous 12 years[1] . Power Conversion Efficiency (PCE) of modern OSCs produced in labs have exceeded 20%, comparable to traditional silicon and perovskite cells2 . Much of the advancement is credited to the development of PM6:Y6 donor:acceptor systems3 . Current challenges in the development of commercially viable materials for OSCs are stability4 and solubility in green solvents for deposition. This proposal is the implementation of the Knoevenagel condensation5 in the preparation of PM6 and Y6 inspired green material libraries and the screening of these libraries for preferred properties of irradiation stability and solubility in green solvents, followed by implementation in bulk heterojunction OSCs to determine PCE.

The Knoevenagel condensation is an aldol condensation that typically forms conjugated systems. This reaction often has the greatest yields in aqueous sodium hydroxide solutions with small quantities of organic solvent present5. This condensation reaction has a much lower environmental impact than the Stille coupling commonly used to synthesize conjugated materials for OSCs. A combinatorial library of polymeric donor materials produced via the Knoevenagel polymerization can be screened by irradiation with light, and extraction with green solvents such as 2-methyl tetrahydrofuran. The surviving polymers can be examined for PCE with multiple NFAs.

2 Background

Background.

2.1 Stille coupling reaction

Tin.

2.2 Previous work

3 Proposed work

3.1 Aim 1

Screening polymeric donor combinatorial libraries for stability by irradiation degradation and green solvent extraction.

3.2 Aim 2

Tin-free synthesis of small molecule and Y6-like NFAs and polymeric PM6-like donor combinatorial libraries by Knoevenagel condensation in water.

3.3 Aim 3

Implementation of optimized materials into bulk heterojunction OSCs to determine PCE.

3.4 Potential challenges and solutions

4 References

(1) Xu, W.; He, W.; Li, G.; Wu, J.; Yang, C.; Cao, Z.; Cheng, P.; Li, H.; Du, Z.; Yu, D. Challenging PM6-like Donor Polymers for Pairing with a Y-Type State-of-the-Art Acceptor in Binary Blends for Bulk Heterojunction Solar Cells. *Phys. Chem. Chem. Phys.* **2023**, *25*, 2916–2925.

5 Budget